

Auteur: Rune Suy
Studierichting: Informatica
Oefeningenreeks 5G nr. 2.1

$$f(x) = -x^2 + 4x$$

$$-x^2 + 4x = 0 \Leftrightarrow x(4 - x) = 0 \Leftrightarrow x = 0 \text{ of } x = 4$$

$$\begin{aligned} V &= \pi \int_0^4 (-x^2 + 4x)^2 dx \\ &= \pi \int_0^4 (x^4 - 8x^3 + 16x^2) dx \\ &= \pi \left[\int_0^4 x^4 dx - 8 \int_0^4 x^3 dx + 16 \int_0^4 x^2 dx \right] \\ &= \pi \left[\left[\frac{x^5}{5} \right]_0^4 - 8 \left[\frac{x^4}{4} \right]_0^4 + 16 \left[\frac{x^3}{3} \right]_0^4 \right] \\ &= \pi \left(\frac{3072}{15} - \frac{7680}{15} + \frac{5120}{15} \right) \\ &= \frac{512\pi}{15} \end{aligned}$$



References